

Práctica Tema 5

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04/05/2025

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1 Modelo base

La productividad del sistema es la productividad del terminal, por lo tanto: $X := \text{MTHRPUT}(\text{TERMINAL})$ (línea 24).

El tiempo de respuesta del sistema se puede calcular a partir de la ley del tiempo de respuesta interactiva: $R = \frac{N}{X} - Z$. Como X ya la hemos calculado, N es el numero total de usuarios y Z el tiempo de servicio del terminal: $R := N / X - \text{MSERVICE}(\text{TERMINAL})$ (línea 27). El tiempo de respuesta de la CPU, $R1$, y el tiempo de respuesta del disco, $R2$, se pueden obtener con la función $\text{MRESPONS}()$ (líneas 25-26).

La cantidad de usuarios en la CPU, $N1$, y la cantidad de usuarios del disco, $N2$, se pueden calcular a partir de la siguiente expresión: $N_i = X_i R_i$ (líneas 30-31).

Ejecutando el modelo se han obtenido los siguientes resultados:

N	R1	R2	R	X	N1	N2
1	0.0300	0.1000	0.9400	0.1119	0.0268	0.0783
2	0.0308	0.1078	1.0010	0.2222	0.0547	0.1677
3	0.0316	0.1168	1.0710	0.3307	0.0837	0.2703
4	0.0325	0.1270	1.1490	0.4372	0.1137	0.3888
5	0.0334	0.1389	1.2390	0.5412	0.1446	0.5261
6	0.0343	0.1526	1.3430	0.6422	0.1764	0.6860
7	0.0352	0.1686	1.4630	0.7398	0.2089	0.8731
8	0.0362	0.1873	1.6010	0.8332	0.2417	1.0920
9	0.0372	0.2092	1.7630	0.9219	0.2747	1.3500
10	0.0382	0.2350	1.9510	1.0050	0.3074	1.6530

```

1 /DECLARE/ QUEUE CPU,DISC,TERMINAL;
2     REAL Z=8.;
3     REAL V_CPU=8.,V_DISC=7.;
4     REAL X,X1,X2,R,R1,R2,N1,N2;
5     INTEGER N;
6 /STATION/ NAME=CPU;
7 &     SCHED=PS;
8     SERVICE=EXP(0.03);
9     TRANSIT=DISC,V_DISC,TERMINAL,1;
10 /STATION/ NAME=DISC;
11     TRANSIT=CPU;
12     SERVICE=EXP(0.1);
13 /STATION/ NAME=TERMINAL;
14     TYPE=INFINITE;
15     INIT=N;
16     SERVICE=EXP(Z);
17     TRANSIT=CPU;
18 /CONTROL/ CLASS=ALL QUEUE;
19 /EXEC/   FOR N:=1 STEP 1 UNTIL 10 DO
20     BEGIN
21         PRINT;
22         PRINT("NUMERO DE USUARIOS=",N);
23         SOLVE;
24
25         X := MTHRPUT(TERMINAL);
26         X1 := MTHRPUT(CPU);
27         X2 := MTHRPUT(DISC);
28         R := N / X - Z;
29         R1 := MRESPONS(CPU);
30         R2 := MRESPONS(DISC);
31         N1 := X1 * R1;
32         N2 := X2 * R2;
33

```

```

34      PRINT("R1 = ", R1);
35      PRINT("R2 = ", R2);
36      PRINT("X = ", X);
37      PRINT("R = ", R);
38      PRINT("N1 = ", N1);
39      PRINT("N2 = ", N2);
40      END;

```

1

```

NUMERO DE USUARIOS=      1
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *         *         *         *         *         *
*CPU       *0.3000E-01*0.2685E-01*0.2685E-01*0.3000E-01*0.8949  *
*          *         *         *         *         *         *
*DISC      *0.1000   *0.7830E-01*0.7830E-01*0.1000   *0.7830  *
*          *         *         *         *         *         *
*TERMINAL  * 8.000   *0.0000E+00*0.8949   * 8.000   *0.1119  *
*          *         *         *         *         *         *
*****
      MEMORY USED:      6992 WORDS OF 4 BYTES
      ( 0.14 % OF TOTAL MEMORY)

R1 = 0.3000E-01
R2 = 0.1000
X = 0.1119
R = 0.9400
N1 = 0.2685E-01
N2 = 0.7830E-01

```

1

```

NUMERO DE USUARIOS=      2
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *         *         *         *         *         *
*CPU       *0.3000E-01*0.5333E-01*0.5476E-01*0.3081E-01* 1.778  *
*          *         *         *         *         *         *
*DISC      *0.1000   *0.1555   *0.1677   *0.1078   * 1.555  *
*          *         *         *         *         *         *
*TERMINAL  * 8.000   *0.0000E+00* 1.778   * 8.000   *0.2222  *
*          *         *         *         *         *         *
*****
      MEMORY USED:      7018 WORDS OF 4 BYTES
      ( 0.14 % OF TOTAL MEMORY)

R1 = 0.3081E-01
R2 = 0.1078
X = 0.2222
R = 1.001
N1 = 0.5476E-01
N2 = 0.1677

```

1

```

NUMERO DE USUARIOS=      3
- MEAN VALUE ANALYSIS ("MVA") -
*****

```

```

*   NAME   *   SERVICE * BUSY PCT *   CUST NB * RESPONSE *   THRUPUT *
*****
*           *           *           *           *           *           *
*CPU       *0.3000E-01*0.7938E-01*0.8372E-01*0.3164E-01* 2.646   *
*           *           *           *           *           *           *
*DISC      *0.1000    *0.2315    *0.2703    *0.1168    * 2.315   *
*           *           *           *           *           *           *
*TERMINAL  * 8.000    *0.0000E+00* 2.646    * 8.000    *0.3307   *
*           *           *           *           *           *           *
*****

```

MEMORY USED: 7044 WORDS OF 4 BYTES
(0.14 % OF TOTAL MEMORY)

R1 = 0.3164E-01
R2 = 0.1168
X = 0.3307
R = 1.071
N1 = 0.8372E-01
N2 = 0.2703

1

NUMERO DE USUARIOS= 4
- MEAN VALUE ANALYSIS ("MVA") -

```

*****
*   NAME   *   SERVICE * BUSY PCT *   CUST NB * RESPONSE *   THRUPUT *
*****
*           *           *           *           *           *           *
*CPU       *0.3000E-01*0.1049    *0.1137    *0.3251E-01* 3.498   *
*           *           *           *           *           *           *
*DISC      *0.1000    *0.3060    *0.3888    *0.1270    * 3.060   *
*           *           *           *           *           *           *
*TERMINAL  * 8.000    *0.0000E+00* 3.498    * 8.000    *0.4372   *
*           *           *           *           *           *           *
*****

```

MEMORY USED: 7070 WORDS OF 4 BYTES
(0.14 % OF TOTAL MEMORY)

R1 = 0.3251E-01
R2 = 0.1270
X = 0.4372
R = 1.149
N1 = 0.1137
N2 = 0.3888

1

NUMERO DE USUARIOS= 5
- MEAN VALUE ANALYSIS ("MVA") -

```

*****
*   NAME   *   SERVICE * BUSY PCT *   CUST NB * RESPONSE *   THRUPUT *
*****
*           *           *           *           *           *           *
*CPU       *0.3000E-01*0.1299    *0.1446    *0.3341E-01* 4.329   *
*           *           *           *           *           *           *
*DISC      *0.1000    *0.3788    *0.5261    *0.1389    * 3.788   *
*           *           *           *           *           *           *
*TERMINAL  * 8.000    *0.0000E+00* 4.329    * 8.000    *0.5412   *
*           *           *           *           *           *           *
*****

```

MEMORY USED: 7096 WORDS OF 4 BYTES
(0.14 % OF TOTAL MEMORY)

```

R1 = 0.3341E-01
R2 = 0.1389
X = 0.5412
R = 1.239
N1 = 0.1446
N2 = 0.5261

```

1

```

NUMERO DE USUARIOS=      6
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.1541  *0.1764  *0.3434E-01* 5.138  *
*          *          *          *          *          *
*DISC      *0.1000   *0.4495   *0.6860  *0.1526   * 4.495  *
*          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00* 5.138  * 8.000   *0.6422  *
*          *          *          *          *          *
*****
      MEMORY USED:      7122 WORDS OF 4 BYTES
      ( 0.14 % OF TOTAL MEMORY)
R1 = 0.3434E-01
R2 = 0.1526
X = 0.6422
R = 1.343
N1 = 0.1764
N2 = 0.6860

```

1

```

NUMERO DE USUARIOS=      7
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.1775  *0.2089  *0.3529E-01* 5.918  *
*          *          *          *          *          *
*DISC      *0.1000   *0.5178   *0.8731  *0.1686   * 5.178  *
*          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00* 5.918  * 8.000   *0.7398  *
*          *          *          *          *          *
*****
      MEMORY USED:      7148 WORDS OF 4 BYTES
      ( 0.14 % OF TOTAL MEMORY)
R1 = 0.3529E-01
R2 = 0.1686
X = 0.7398
R = 1.463
N1 = 0.2089
N2 = 0.8731

```

1

```

NUMERO DE USUARIOS=      8
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****

```

```
*****
*           *           *           *           *           *
*CPU      *0.3000E-01*0.2000    *0.2417    *0.3627E-01* 6.666 *
*           *           *           *           *           *
*DISC      *0.1000    *0.5833    * 1.092    *0.1873    * 5.833 *
*           *           *           *           *           *
*TERMINAL * 8.000    *0.0000E+00* 6.666    * 8.000    *0.8332 *
*           *           *           *           *           *
*****
      MEMORY USED:      7174 WORDS OF 4 BYTES
      ( 0.14 % OF TOTAL MEMORY)

R1 = 0.3627E-01
R2 = 0.1873
X = 0.8332
R = 1.601
N1 = 0.2417
N2 = 1.092
```

```
1
NUMERO DE USUARIOS=      9
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME  *  SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*           *           *           *           *           *
*CPU      *0.3000E-01*0.2212    *0.2747    *0.3725E-01* 7.375 *
*           *           *           *           *           *
*DISC      *0.1000    *0.6453    * 1.350    *0.2092    * 6.453 *
*           *           *           *           *           *
*TERMINAL * 8.000    *0.0000E+00* 7.375    * 8.000    *0.9219 *
*           *           *           *           *           *
*****
      MEMORY USED:      7200 WORDS OF 4 BYTES
      ( 0.14 % OF TOTAL MEMORY)

R1 = 0.3725E-01
R2 = 0.2092
X = 0.9219
R = 1.763
N1 = 0.2747
N2 = 1.350
```

```
1
NUMERO DE USUARIOS=     10
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME  *  SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*           *           *           *           *           *
*CPU      *0.3000E-01*0.2412    *0.3074    *0.3824E-01* 8.039 *
*           *           *           *           *           *
*DISC      *0.1000    *0.7034    * 1.653    *0.2350    * 7.034 *
*           *           *           *           *           *
*TERMINAL * 8.000    *0.0000E+00* 8.039    * 8.000    * 1.005 *
*           *           *           *           *           *
*****
      MEMORY USED:      7226 WORDS OF 4 BYTES
      ( 0.14 % OF TOTAL MEMORY)

R1 = 0.3824E-01
```

```
R2 = 0.2350
X = 1.005
R = 1.951
N1 = 0.3074
N2 = 1.653
41 /END/
```

2 Apartado a

Las demandas de cada dispositivo se calculan con la expresión: $D_i = V_i \cdot S_i$ (líneas 34-35).

$$D_1 = V_1 \cdot S_2 = 8 \cdot 0.03 = 0.24$$

$$D_2 = V_2 \cdot S_2 = 7 \cdot 0.1 = 0.7$$

La demanda total es la suma de todas las demandas: $D = \sum_{i=1}^K D_i = D_1 + D_2 = 0.24 + 0.7 = 0.94$ (línea 36).

La demanda del cuello de botella es la demanda máxima de todos los dispositivos: $D_b = \max_{i=1 \dots K} \{D_i\} = \max\{D_1, D_2\} = \max\{0.24, 0.7\} = 0.7 = D_2$ (línea 37).

El punto de saturación: $N^* = \left\lceil \frac{D+Z}{D_b} \right\rceil = \left\lceil \frac{0.94+8}{0.7} \right\rceil = \lceil 12.7714 \rceil = 13$ (línea 38).

```

1 /DECLARE/ QUEUE CPU,DISC,TERMINAL;
2     REAL Z=8.;
3     REAL V_CPU=8.,V_DISC=7.;
4     REAL X,X1,X2,R,R1,R2,N1,N2;
5     REAL D,D1,D2,Db,Ns;
6     INTEGER N;
7 /STATION/ NAME=CPU;
8 &
9     SCHED=PS;
10    SERVICE=EXP(0.03);
11    TRANSIT=DISC,V_DISC,TERMINAL,1;
12 /STATION/ NAME=DISC;
13    TRANSIT=CPU;
14    SERVICE=EXP(0.1);
15 /STATION/ NAME=TERMINAL;
16    TYPE=INFINITE;
17    INIT=N;
18    SERVICE=EXP(Z);
19    TRANSIT=CPU;
20 /CONTROL/ CLASS=ALL QUEUE;
21 /EXEC/    FOR N:=1 STEP 1 UNTIL 10 DO
22    BEGIN
23        PRINT;
24        PRINT("NUMERO DE USUARIOS=",N);
25        SOLVE;
26
27        X := MTHRUPUT(TERMINAL);
28        X1 := MTHRUPUT(CPU);
29        X2 := MTHRUPUT(DISC);
30        R := N / X - Z;
31        R1 := MRESPONS(CPU);
32        R2 := MRESPONS(DISC);
33        N1 := X1 * R1;
34        N2 := X2 * R2;
35        D1 := MSERVICE(CPU) * V_CPU;
36        D2 := MSERVICE(DISC) * V_DISC;
37        D := D1 + D2;
38        Db := MAX(D1,D2);
39        Ns := INTROUND((D + Z) / Db);
40
41        PRINT("R1 = ", R1);
42        PRINT("R2 = ", R2);
43        PRINT("X = ", X);
44        PRINT("R = ", R);
45        PRINT("N1 = ", N1);

```

```

45      PRINT("N2 = ", N2);
46      PRINT("D1 = ", D1);
47      PRINT("D2 = ", D2);
48      PRINT("D  = ", D);
49      PRINT("Db = ", Db);
50      PRINT("Ns = ", Ns);
51      END;

```

1

```

NUMERO DE USUARIOS=      1
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*          *         *          *          *          *          *
*CPU       *0.3000E-01*0.2685E-01*0.2685E-01*0.3000E-01*0.8949  *
*          *         *          *          *          *          *
*DISC      *0.1000   *0.7830E-01*0.7830E-01*0.1000   *0.7830  *
*          *         *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00*0.8949   * 8.000   *0.1119  *
*          *         *          *          *          *          *
*****
      MEMORY USED:      7158 WORDS OF 4 BYTES
      ( 0.14 % OF TOTAL MEMORY)

R1 =  0.3000E-01
R2 =  0.1000
X =  0.1119
R =  0.9400
N1 =  0.2685E-01
N2 =  0.7830E-01
D1 =  0.2400
D2 =  0.7000
D =  0.9400
Db =  0.7000
Ns =  13.00

```

1

```

NUMERO DE USUARIOS=      2
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*          *         *          *          *          *          *
*CPU       *0.3000E-01*0.5333E-01*0.5476E-01*0.3081E-01* 1.778  *
*          *         *          *          *          *          *
*DISC      *0.1000   *0.1555   *0.1677   *0.1078   * 1.555  *
*          *         *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00* 1.778   * 8.000   *0.2222  *
*          *         *          *          *          *          *
*****
      MEMORY USED:      7184 WORDS OF 4 BYTES
      ( 0.14 % OF TOTAL MEMORY)

R1 =  0.3081E-01
R2 =  0.1078
X =  0.2222
R =  1.001
N1 =  0.5476E-01
N2 =  0.1677

```

D1 = 0.2400
 D2 = 0.7000
 D = 0.9400
 Db = 0.7000
 Ns = 13.00

1

```

NUMERO DE USUARIOS=      3
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *         *         *         *         *
*CPU       *0.3000E-01*0.7938E-01*0.8372E-01*0.3164E-01* 2.646 *
*          *         *         *         *         *
*DISC      *0.1000   *0.2315   *0.2703   *0.1168   * 2.315 *
*          *         *         *         *         *
*TERMINAL  * 8.000   *0.0000E+00* 2.646   * 8.000   *0.3307 *
*          *         *         *         *         *
*****
          MEMORY USED:      7210 WORDS OF 4 BYTES
          ( 0.14 % OF TOTAL MEMORY)

R1 = 0.3164E-01
R2 = 0.1168
X = 0.3307
R = 1.071
N1 = 0.8372E-01
N2 = 0.2703
D1 = 0.2400
D2 = 0.7000
D = 0.9400
Db = 0.7000
Ns = 13.00

```

1

```

NUMERO DE USUARIOS=      4
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *         *         *         *         *
*CPU       *0.3000E-01*0.1049   *0.1137   *0.3251E-01* 3.498 *
*          *         *         *         *         *
*DISC      *0.1000   *0.3060   *0.3888   *0.1270   * 3.060 *
*          *         *         *         *         *
*TERMINAL  * 8.000   *0.0000E+00* 3.498   * 8.000   *0.4372 *
*          *         *         *         *         *
*****
          MEMORY USED:      7236 WORDS OF 4 BYTES
          ( 0.14 % OF TOTAL MEMORY)

R1 = 0.3251E-01
R2 = 0.1270
X = 0.4372
R = 1.149
N1 = 0.1137
N2 = 0.3888
D1 = 0.2400
D2 = 0.7000

```

D = 0.9400
 Db = 0.7000
 Ns = 13.00

1

```

NUMERO DE USUARIOS=      5
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.1299   *0.1446   *0.3341E-01* 4.329 *
*          *          *          *          *          *
*DISC      *0.1000   *0.3788   *0.5261   *0.1389   * 3.788 *
*          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00* 4.329   * 8.000   *0.5412 *
*          *          *          *          *          *
*****
                MEMORY USED:      7262 WORDS OF 4 BYTES
                ( 0.15 % OF TOTAL MEMORY)

R1 = 0.3341E-01
R2 = 0.1389
X = 0.5412
R = 1.239
N1 = 0.1446
N2 = 0.5261
D1 = 0.2400
D2 = 0.7000
D = 0.9400
Db = 0.7000
Ns = 13.00

```

1

```

NUMERO DE USUARIOS=      6
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.1541   *0.1764   *0.3434E-01* 5.138 *
*          *          *          *          *          *
*DISC      *0.1000   *0.4495   *0.6860   *0.1526   * 4.495 *
*          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00* 5.138   * 8.000   *0.6422 *
*          *          *          *          *          *
*****
                MEMORY USED:      7288 WORDS OF 4 BYTES
                ( 0.15 % OF TOTAL MEMORY)

R1 = 0.3434E-01
R2 = 0.1526
X = 0.6422
R = 1.343
N1 = 0.1764
N2 = 0.6860
D1 = 0.2400
D2 = 0.7000
D = 0.9400
Db = 0.7000

```

Ns = 13.00

1

```

NUMERO DE USUARIOS=      7
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *         *         *         *         *         *
*CPU       *0.3000E-01*0.1775  *0.2089  *0.3529E-01* 5.918  *
*          *         *         *         *         *         *
*DISC      *0.1000    *0.5178  *0.8731  *0.1686  * 5.178  *
*          *         *         *         *         *         *
*TERMINAL  * 8.000    *0.0000E+00* 5.918  * 8.000  *0.7398  *
*          *         *         *         *         *         *
*****
          MEMORY USED:      7314 WORDS OF 4 BYTES
          ( 0.15 % OF TOTAL MEMORY)

R1 = 0.3529E-01
R2 = 0.1686
X = 0.7398
R = 1.463
N1 = 0.2089
N2 = 0.8731
D1 = 0.2400
D2 = 0.7000
D = 0.9400
Db = 0.7000
Ns = 13.00

```

1

```

NUMERO DE USUARIOS=      8
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *         *         *         *         *         *
*CPU       *0.3000E-01*0.2000  *0.2417  *0.3627E-01* 6.666  *
*          *         *         *         *         *         *
*DISC      *0.1000    *0.5833  * 1.092  *0.1873  * 5.833  *
*          *         *         *         *         *         *
*TERMINAL  * 8.000    *0.0000E+00* 6.666  * 8.000  *0.8332  *
*          *         *         *         *         *         *
*****
          MEMORY USED:      7340 WORDS OF 4 BYTES
          ( 0.15 % OF TOTAL MEMORY)

R1 = 0.3627E-01
R2 = 0.1873
X = 0.8332
R = 1.601
N1 = 0.2417
N2 = 1.092
D1 = 0.2400
D2 = 0.7000
D = 0.9400
Db = 0.7000
Ns = 13.00

```

```

1
NUMERO DE USUARIOS=          9
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *          *
*CPU       *0.3000E-01*0.2212   *0.2747   *0.3725E-01*  7.375   *
*          *          *          *          *          *          *
*DISC      *0.1000   *0.6453   *  1.350   *0.2092   *  6.453   *
*          *          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00*  7.375   * 8.000   *0.9219   *
*          *          *          *          *          *          *
*****
          MEMORY USED:      7366 WORDS OF 4 BYTES
          ( 0.15  % OF TOTAL MEMORY)

R1 =   0.3725E-01
R2 =   0.2092
X =   0.9219
R =   1.763
N1 =   0.2747
N2 =   1.350
D1 =   0.2400
D2 =   0.7000
D =   0.9400
Db =   0.7000
Ns =   13.00

```

```

1
NUMERO DE USUARIOS=         10
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *          *
*CPU       *0.3000E-01*0.2412   *0.3074   *0.3824E-01*  8.039   *
*          *          *          *          *          *          *
*DISC      *0.1000   *0.7034   *  1.653   *0.2350   *  7.034   *
*          *          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00*  8.039   * 8.000   *  1.005   *
*          *          *          *          *          *          *
*****
          MEMORY USED:      7392 WORDS OF 4 BYTES
          ( 0.15  % OF TOTAL MEMORY)

R1 =   0.3824E-01
R2 =   0.2350
X =   1.005
R =   1.951
N1 =   0.3074
N2 =   1.653
D1 =   0.2400
D2 =   0.7000
D =   0.9400
Db =   0.7000
Ns =   13.00
52 /END/

```

3 Apartado b

El tiempo de respuesta del sistema, R , ya lo tenemos calculado a partir del modelo base (línea 30).

El tiempo de respuesta total se calcula: $R_t = R + Z$, (línea 40).

Para la cantidad de usuarios que están trabajando $N_w = X_0 \cdot R$ (línea 41), y están pensando $N_z = X_0 \cdot Z$ (línea 42).

Ejecutando el modelo obtenemos los siguientes resultados:

N	R1	R2	R	X	N1	N2	Rt	Nw	Nz
1	0.0300	0.1000	0.9400	0.1119	0.0268	0.0783	8.9400	0.1051	0.8949
2	0.0308	0.1078	1.0010	0.2222	0.0547	0.1677	9.0010	0.2225	1.7780
3	0.0316	0.1168	1.0710	0.3307	0.0837	0.2703	9.0710	0.3541	2.6460
4	0.0325	0.1270	1.1490	0.4372	0.1137	0.3888	9.1490	0.5025	3.4980
5	0.0334	0.1389	1.2390	0.5412	0.1446	0.5261	9.2390	0.6707	4.3290
6	0.0343	0.1526	1.3430	0.6422	0.1764	0.6860	9.3430	0.8624	5.1380
7	0.0352	0.1686	1.4630	0.7398	0.2089	0.8731	9.4630	1.0820	5.9180
8	0.0362	0.1873	1.6010	0.8332	0.2417	1.0920	9.6010	1.3340	6.6660
9	0.0372	0.2092	1.7630	0.9219	0.2747	1.3500	9.7630	1.6250	7.3750
10	0.0382	0.2350	1.9510	1.0050	0.3074	1.6530	9.9510	1.9610	8.0390

```

1 /DECLARE/ QUEUE CPU,DISC,TERMINAL;
2     REAL Z=8.;
3     REAL V_CPU=8.,V_DISC=7.;
4     REAL X,X1,X2,R,R1,R2,N1,N2;
5     REAL D,D1,D2,Db,Ns;
6     REAL Rt,Nw,Nz;
7     INTEGER N;
8 /STATION/ NAME=CPU;
9 &     SCHED=PS;
10     SERVICE=EXP(0.03);
11     TRANSIT=DISC,V_DISC,TERMINAL,1;
12 /STATION/ NAME=DISC;
13     TRANSIT=CPU;
14     SERVICE=EXP(0.1);
15 /STATION/ NAME=TERMINAL;
16     TYPE=INFINITE;
17     INIT=N;
18     SERVICE=EXP(Z);
19     TRANSIT=CPU;
20 /CONTROL/ CLASS=ALL QUEUE;
21 /EXEC/   FOR N:=1 STEP 1 UNTIL 10 DO
22     BEGIN
23         PRINT;
24         PRINT("NUMERO DE USUARIOS=",N);
25         SOLVE;
26
27         X := MTHRUPUT(TERMINAL);
28         X1 := MTHRUPUT(CPU);
29         X2 := MTHRUPUT(DISC);
30         R := N / X - Z;
31         R1 := MRESPONS(CPU);
32         R2 := MRESPONS(DISC);
33         N1 := X1 * R1;
34         N2 := X2 * R2;
35         D1 := MSERVICE(CPU) * V_CPU;
36         D2 := MSERVICE(DISC) * V_DISC;
37         D := D1 + D2;

```

```

38      Db := MAX(D1,D2);
39      Ns := INTROUND((D + Z) / Db);
40      Rt := R + Z;
41      Nw := X * R;
42      Nz := X * Z;
43
44      PRINT("R1 = ", R1);
45      PRINT("R2 = ", R2);
46      PRINT("X = ", X);
47      PRINT("R = ", R);
48      PRINT("N1 = ", N1);
49      PRINT("N2 = ", N2);
50      PRINT("D1 = ", D1);
51      PRINT("D2 = ", D2);
52      PRINT("D = ", D);
53      PRINT("Db = ", Db);
54      PRINT("Ns = ", Ns);
55      PRINT("Rt = ", Rt);
56      PRINT("Nw = ", Nw);
57      PRINT("Nz = ", Nz);
58      END;

```

1

```

NUMERO DE USUARIOS=      1
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.2685E-01*0.2685E-01*0.3000E-01*0.8949 *
*          *          *          *          *          *
*DISC      *0.1000   *0.7830E-01*0.7830E-01*0.1000   *0.7830 *
*          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00*0.8949   * 8.000   *0.1119 *
*          *          *          *          *          *
*****
      MEMORY USED:      7248 WORDS OF 4 BYTES
      ( 0.14 % OF TOTAL MEMORY)

R1 =  0.3000E-01
R2 =  0.1000
X =  0.1119
R =  0.9400
N1 =  0.2685E-01
N2 =  0.7830E-01
D1 =  0.2400
D2 =  0.7000
D =  0.9400
Db =  0.7000
Ns =  13.00
Rt =  8.940
Nw =  0.1051
Nz =  0.8949

```

1

```

NUMERO DE USUARIOS=      2
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****

```

```
*****
*           *           *           *           *           *
*CPU      *0.3000E-01*0.5333E-01*0.5476E-01*0.3081E-01* 1.778 *
*           *           *           *           *           *
*DISC     *0.1000    *0.1555    *0.1677    *0.1078    * 1.555 *
*           *           *           *           *           *
*TERMINAL * 8.000    *0.0000E+00* 1.778    * 8.000    *0.2222 *
*           *           *           *           *           *
*****
```

MEMORY USED: 7274 WORDS OF 4 BYTES
(0.15 % OF TOTAL MEMORY)

```
R1 = 0.3081E-01
R2 = 0.1078
X = 0.2222
R = 1.001
N1 = 0.5476E-01
N2 = 0.1677
D1 = 0.2400
D2 = 0.7000
D = 0.9400
Db = 0.7000
Ns = 13.00
Rt = 9.001
Nw = 0.2225
Nz = 1.778
```

1

NUMERO DE USUARIOS= 3
- MEAN VALUE ANALYSIS ("MVA") -

```
*****
* NAME * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*           *           *           *           *           *
*CPU      *0.3000E-01*0.7938E-01*0.8372E-01*0.3164E-01* 2.646 *
*           *           *           *           *           *
*DISC     *0.1000    *0.2315    *0.2703    *0.1168    * 2.315 *
*           *           *           *           *           *
*TERMINAL * 8.000    *0.0000E+00* 2.646    * 8.000    *0.3307 *
*           *           *           *           *           *
*****
```

MEMORY USED: 7300 WORDS OF 4 BYTES
(0.15 % OF TOTAL MEMORY)

```
R1 = 0.3164E-01
R2 = 0.1168
X = 0.3307
R = 1.071
N1 = 0.8372E-01
N2 = 0.2703
D1 = 0.2400
D2 = 0.7000
D = 0.9400
Db = 0.7000
Ns = 13.00
Rt = 9.071
Nw = 0.3541
Nz = 2.646
```

1

```

NUMERO DE USUARIOS=      4
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *          *
*CPU       *0.3000E-01*0.1049   *0.1137   *0.3251E-01*  3.498   *
*          *          *          *          *          *          *
*DISC      *0.1000    *0.3060   *0.3888   *0.1270   *  3.060   *
*          *          *          *          *          *          *
*TERMINAL  * 8.000    *0.0000E+00*  3.498   *  8.000   *0.4372   *
*          *          *          *          *          *          *
*****
          MEMORY USED:      7326 WORDS OF 4 BYTES
          ( 0.15  % OF TOTAL MEMORY)

R1 =   0.3251E-01
R2 =   0.1270
X =   0.4372
R =    1.149
N1 =   0.1137
N2 =   0.3888
D1 =   0.2400
D2 =   0.7000
D =   0.9400
Db =   0.7000
Ns =   13.00
Rt =    9.149
Nw =   0.5025
Nz =    3.498

```

1

```

NUMERO DE USUARIOS=      5
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *          *
*CPU       *0.3000E-01*0.1299   *0.1446   *0.3341E-01*  4.329   *
*          *          *          *          *          *          *
*DISC      *0.1000    *0.3788   *0.5261   *0.1389   *  3.788   *
*          *          *          *          *          *          *
*TERMINAL  * 8.000    *0.0000E+00*  4.329   *  8.000   *0.5412   *
*          *          *          *          *          *          *
*****
          MEMORY USED:      7352 WORDS OF 4 BYTES
          ( 0.15  % OF TOTAL MEMORY)

R1 =   0.3341E-01
R2 =   0.1389
X =   0.5412
R =    1.239
N1 =   0.1446
N2 =   0.5261
D1 =   0.2400
D2 =   0.7000
D =   0.9400
Db =   0.7000
Ns =   13.00
Rt =    9.239

```

Nw = 0.6707

Nz = 4.329

1

NUMERO DE USUARIOS= 6

- MEAN VALUE ANALYSIS ("MVA") -

```
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.1541  *0.1764  *0.3434E-01* 5.138  *
*          *          *          *          *          *
*DISC      *0.1000    *0.4495  *0.6860  *0.1526  * 4.495  *
*          *          *          *          *          *
*TERMINAL  * 8.000    *0.0000E+00* 5.138  * 8.000  *0.6422  *
*          *          *          *          *          *
```

MEMORY USED: 7378 WORDS OF 4 BYTES

(0.15 % OF TOTAL MEMORY)

R1 = 0.3434E-01

R2 = 0.1526

X = 0.6422

R = 1.343

N1 = 0.1764

N2 = 0.6860

D1 = 0.2400

D2 = 0.7000

D = 0.9400

Db = 0.7000

Ns = 13.00

Rt = 9.343

Nw = 0.8624

Nz = 5.138

1

NUMERO DE USUARIOS= 7

- MEAN VALUE ANALYSIS ("MVA") -

```
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.1775  *0.2089  *0.3529E-01* 5.918  *
*          *          *          *          *          *
*DISC      *0.1000    *0.5178  *0.8731  *0.1686  * 5.178  *
*          *          *          *          *          *
*TERMINAL  * 8.000    *0.0000E+00* 5.918  * 8.000  *0.7398  *
*          *          *          *          *          *
```

MEMORY USED: 7404 WORDS OF 4 BYTES

(0.15 % OF TOTAL MEMORY)

R1 = 0.3529E-01

R2 = 0.1686

X = 0.7398

R = 1.463

N1 = 0.2089

N2 = 0.8731

D1 = 0.2400

D2 = 0.7000

D = 0.9400
 Db = 0.7000
 Ns = 13.00
 Rt = 9.463
 Nw = 1.082
 Nz = 5.918

1

NUMERO DE USUARIOS= 8
 - MEAN VALUE ANALYSIS ("MVA") -

 * NAME * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *

 * * * * *
 *CPU *0.3000E-01*0.2000 *0.2417 *0.3627E-01* 6.666 *
 * * * * *
 *DISC *0.1000 *0.5833 * 1.092 *0.1873 * 5.833 *
 * * * * *
 *TERMINAL * 8.000 *0.0000E+00* 6.666 * 8.000 *0.8332 *
 * * * * *

 MEMORY USED: 7430 WORDS OF 4 BYTES
 (0.15 % OF TOTAL MEMORY)
 R1 = 0.3627E-01
 R2 = 0.1873
 X = 0.8332
 R = 1.601
 N1 = 0.2417
 N2 = 1.092
 D1 = 0.2400
 D2 = 0.7000
 D = 0.9400
 Db = 0.7000
 Ns = 13.00
 Rt = 9.601
 Nw = 1.334
 Nz = 6.666

1

NUMERO DE USUARIOS= 9
 - MEAN VALUE ANALYSIS ("MVA") -

 * NAME * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *

 * * * * *
 *CPU *0.3000E-01*0.2212 *0.2747 *0.3725E-01* 7.375 *
 * * * * *
 *DISC *0.1000 *0.6453 * 1.350 *0.2092 * 6.453 *
 * * * * *
 *TERMINAL * 8.000 *0.0000E+00* 7.375 * 8.000 *0.9219 *
 * * * * *

 MEMORY USED: 7456 WORDS OF 4 BYTES
 (0.15 % OF TOTAL MEMORY)
 R1 = 0.3725E-01
 R2 = 0.2092
 X = 0.9219
 R = 1.763

N1 = 0.2747
 N2 = 1.350
 D1 = 0.2400
 D2 = 0.7000
 D = 0.9400
 Db = 0.7000
 Ns = 13.00
 Rt = 9.763
 Nw = 1.625
 Nz = 7.375

1

```

NUMERO DE USUARIOS=      10
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.2412   *0.3074   *0.3824E-01* 8.039   *
*          *          *          *          *          *
*DISC      *0.1000   *0.7034   * 1.653   *0.2350   * 7.034   *
*          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00* 8.039   * 8.000   * 1.005   *
*          *          *          *          *          *
*****
          MEMORY USED:      7482 WORDS OF 4 BYTES
          ( 0.15  % OF TOTAL MEMORY)

R1 = 0.3824E-01
R2 = 0.2350
X = 1.005
R = 1.951
N1 = 0.3074
N2 = 1.653
D1 = 0.2400
D2 = 0.7000
D = 0.9400
Db = 0.7000
Ns = 13.00
Rt = 9.951
Nw = 1.961
Nz = 8.039
59 /END/
  
```

4 Apartado c

Se ha cambiado DISC por un vector de 2 entradas DISC(2) (línea 1), se ha dividido en partes iguales la razón de visita del disco original entre los dos gemelos (línea 3), se ha duplicado la estación DISC con configuraciones idénticas (líneas 10-15) y se ha añadido el cálculo del tiempo de respuesta y cantidad de trabajos para el nuevo disco (líneas 35 y 38).

Ejecutando el modelo obtenemos los siguientes resultados:

N	R1	R2	R3	R	X	N1	N2	N3
1	0.0300	0.1000	0.1000	0.1119	0.9400	0.0268	0.0391	0.0391
2	0.0308	0.1039	0.1039	0.2229	0.9738	0.0549	0.0810	0.0810
3	0.0316	0.1081	0.1081	0.3330	1.0100	0.0843	0.1260	0.1260
4	0.0325	0.1126	0.1126	0.4421	1.0480	0.1150	0.1742	0.1742
5	0.0334	0.1174	0.1174	0.5501	1.0900	0.1472	0.2261	0.2261
6	0.0344	0.1226	0.1226	0.6569	1.1340	0.1809	0.2819	0.2819
7	0.0354	0.1282	0.1282	0.7625	1.1810	0.2161	0.3421	0.3421
8	0.0364	0.1342	0.1342	0.8666	1.2310	0.2529	0.4071	0.4071
9	0.0375	0.1407	0.1407	0.9692	1.2860	0.2915	0.4773	0.4773
10	0.0387	0.1477	0.1477	1.0700	1.3440	0.3317	0.5534	0.5534

Como era de esperar el tiempo de respuesta y el número de trabajos en cada disco son idénticos entre ambos discos.

El tiempo de respuesta de los discos ha disminuido debido a que la cantidad de trabajos también lo ha hecho al repartirse la carga entre dos discos en lugar de uno. Como el disco era el cuello de botella en el apartado anterior y ahora hemos mejorado su rendimiento el tiempo de respuesta del sistema también ha disminuido.

```

1 /DECLARE/ QUEUE CPU,DISC(2),TERMINAL;
2     REAL Z=8.;
3     REAL V_CPU=8.,V_DISC(2)=(3.5,3.5);
4     REAL X,X1,X2,X3,R,R1,R2,R3,N1,N2,N3;
5     INTEGER N;
6 /STATION/ NAME=CPU;
7 &     SCHED=PS;
8     SERVICE=EXP(0.03);
9     TRANSIT=DISC,V_DISC,TERMINAL,1;
10 /STATION/ NAME=DISC;
11     TRANSIT=CPU;
12 /STATION/ NAME=DISC(1);
13     SERVICE=EXP(0.1);
14 /STATION/ NAME=DISC(2);
15     SERVICE=EXP(0.1);
16 /STATION/ NAME=TERMINAL;
17     TYPE=INFINITE;
18     INIT=N;
19     SERVICE=EXP(Z);
20     TRANSIT=CPU;
21 /CONTROL/ CLASS=ALL QUEUE;
22 /EXEC/   FOR N:=1 STEP 1 UNTIL 10 DO
23     BEGIN
24         PRINT;
25         PRINT("NUMERO DE USUARIOS=",N);
26         SOLVE;
27
28         X := MTHRUPUT(TERMINAL);
29         X1 := MTHRUPUT(CPU);
30         X2 := MTHRUPUT(DISC(1));
31         X3 := MTHRUPUT(DISC(2));
32         R := N / X - Z;
```

```

33      R1 := MRESPONS(CPU);
34      R2 := MRESPONS(DISC(1));
35      R3 := MRESPONS(DISC(2));
36      N1 := X1 * R1;
37      N2 := X2 * R2;
38      N3 := X2 * R3;
39
40      PRINT("R1 = ", R1);
41      PRINT("R2 = ", R2);
42      PRINT("R3 = ", R3);
43      PRINT("X = ", X);
44      PRINT("R  = ", R);
45      PRINT("N1 = ", N1);
46      PRINT("N2 = ", N2);
47      PRINT("N3 = ", N3);
48      END;

```

1

```

NUMERO DE USUARIOS=      1
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.2685E-01*0.2685E-01*0.3000E-01*0.8949 *
*          *          *          *          *          *
*DISC  1   *0.1000   *0.3915E-01*0.3915E-01*0.1000   *0.3915 *
*          *          *          *          *          *
*DISC  2   *0.1000   *0.3915E-01*0.3915E-01*0.1000   *0.3915 *
*          *          *          *          *          *
*TERMINAL * 8.000    *0.0000E+00*0.8949   * 8.000    *0.1119 *
*          *          *          *          *          *
*****
      MEMORY USED:      7394 WORDS OF 4 BYTES
      ( 0.15 % OF TOTAL MEMORY)

R1 =  0.3000E-01
R2 =  0.1000
R3 =  0.1000
X =  0.1119
R =  0.9400
N1 =  0.2685E-01
N2 =  0.3915E-01
N3 =  0.3915E-01

```

1

```

NUMERO DE USUARIOS=      2
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.5349E-01*0.5492E-01*0.3081E-01* 1.783 *
*          *          *          *          *          *
*DISC  1   *0.1000   *0.7800E-01*0.8106E-01*0.1039   *0.7800 *
*          *          *          *          *          *
*DISC  2   *0.1000   *0.7800E-01*0.8106E-01*0.1039   *0.7800 *
*          *          *          *          *          *
*TERMINAL * 8.000    *0.0000E+00* 1.783   * 8.000    *0.2229 *

```

```

*           *           *           *           *           *           *
*****
      MEMORY USED:      7417 WORDS OF 4 BYTES
      ( 0.15 % OF TOTAL MEMORY)
R1 = 0.3081E-01
R2 = 0.1039
R3 = 0.1039
X = 0.2229
R = 0.9738
N1 = 0.5492E-01
N2 = 0.8106E-01
N3 = 0.8106E-01

```

```

1
NUMERO DE USUARIOS=      3
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME  *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*           *           *           *           *           *           *
*CPU      *0.3000E-01*0.7991E-01*0.8430E-01*0.3165E-01* 2.664 *
*           *           *           *           *           *           *
*DISC  1  *0.1000   *0.1165   *0.1260   *0.1081   * 1.165 *
*           *           *           *           *           *           *
*DISC  2  *0.1000   *0.1165   *0.1260   *0.1081   * 1.165 *
*           *           *           *           *           *           *
*TERMINAL * 8.000    *0.0000E+00* 2.664   * 8.000    *0.3330 *
*           *           *           *           *           *           *
*****
      MEMORY USED:      7440 WORDS OF 4 BYTES
      ( 0.15 % OF TOTAL MEMORY)
R1 = 0.3165E-01
R2 = 0.1081
R3 = 0.1081
X = 0.3330
R = 1.010
N1 = 0.8430E-01
N2 = 0.1260
N3 = 0.1260

```

```

1
NUMERO DE USUARIOS=      4
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME  *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*           *           *           *           *           *           *
*CPU      *0.3000E-01*0.1061   *0.1150   *0.3253E-01* 3.537 *
*           *           *           *           *           *           *
*DISC  1  *0.1000   *0.1547   *0.1742   *0.1126   * 1.547 *
*           *           *           *           *           *           *
*DISC  2  *0.1000   *0.1547   *0.1742   *0.1126   * 1.547 *
*           *           *           *           *           *           *
*TERMINAL * 8.000    *0.0000E+00* 3.537   * 8.000    *0.4421 *
*           *           *           *           *           *           *
*****
      MEMORY USED:      7463 WORDS OF 4 BYTES
      ( 0.15 % OF TOTAL MEMORY)

```

R1 = 0.3253E-01
 R2 = 0.1126
 R3 = 0.1126
 X = 0.4421
 R = 1.048
 N1 = 0.1150
 N2 = 0.1742
 N3 = 0.1742

1

NUMERO DE USUARIOS= 5
 - MEAN VALUE ANALYSIS ("MVA") -

* NAME	* SERVICE	* BUSY PCT	* CUST NB	* RESPONSE	* THRUPUT
*CPU	*0.3000E-01	*0.1320	*0.1472	*0.3345E-01	* 4.401
*DISC 1	*0.1000	*0.1925	*0.2261	*0.1174	* 1.925
*DISC 2	*0.1000	*0.1925	*0.2261	*0.1174	* 1.925
*TERMINAL	* 8.000	*0.0000E+00	* 4.401	* 8.000	*0.5501

 MEMORY USED: 7486 WORDS OF 4 BYTES
 (0.15 % OF TOTAL MEMORY)

R1 = 0.3345E-01
 R2 = 0.1174
 R3 = 0.1174
 X = 0.5501
 R = 1.090
 N1 = 0.1472
 N2 = 0.2261
 N3 = 0.2261

1

NUMERO DE USUARIOS= 6
 - MEAN VALUE ANALYSIS ("MVA") -

* NAME	* SERVICE	* BUSY PCT	* CUST NB	* RESPONSE	* THRUPUT
*CPU	*0.3000E-01	*0.1577	*0.1809	*0.3442E-01	* 5.255
*DISC 1	*0.1000	*0.2299	*0.2819	*0.1226	* 2.299
*DISC 2	*0.1000	*0.2299	*0.2819	*0.1226	* 2.299
*TERMINAL	* 8.000	*0.0000E+00	* 5.255	* 8.000	*0.6569

 MEMORY USED: 7509 WORDS OF 4 BYTES
 (0.15 % OF TOTAL MEMORY)

R1 = 0.3442E-01
 R2 = 0.1226
 R3 = 0.1226
 X = 0.6569

R = 1.134
 N1 = 0.1809
 N2 = 0.2819
 N3 = 0.2819

1

```

NUMERO DE USUARIOS=      7
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.1830   *0.2161   *0.3543E-01* 6.100   *
*          *          *          *          *          *
*DISC  1   *0.1000   *0.2669   *0.3421   *0.1282   * 2.669   *
*          *          *          *          *          *
*DISC  2   *0.1000   *0.2669   *0.3421   *0.1282   * 2.669   *
*          *          *          *          *          *
*TERMINAL * 8.000    *0.0000E+00* 6.100   * 8.000    *0.7625   *
*          *          *          *          *          *
*****
      MEMORY USED:      7532 WORDS OF 4 BYTES
      ( 0.15 % OF TOTAL MEMORY)
R1 = 0.3543E-01
R2 = 0.1282
R3 = 0.1282
X = 0.7625
R = 1.181
N1 = 0.2161
N2 = 0.3421
N3 = 0.3421

```

1

```

NUMERO DE USUARIOS=      8
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.2080   *0.2529   *0.3648E-01* 6.933   *
*          *          *          *          *          *
*DISC  1   *0.1000   *0.3033   *0.4071   *0.1342   * 3.033   *
*          *          *          *          *          *
*DISC  2   *0.1000   *0.3033   *0.4071   *0.1342   * 3.033   *
*          *          *          *          *          *
*TERMINAL * 8.000    *0.0000E+00* 6.933   * 8.000    *0.8666   *
*          *          *          *          *          *
*****
      MEMORY USED:      7555 WORDS OF 4 BYTES
      ( 0.15 % OF TOTAL MEMORY)
R1 = 0.3648E-01
R2 = 0.1342
R3 = 0.1342
X = 0.8666
R = 1.231
N1 = 0.2529
N2 = 0.4071
N3 = 0.4071

```

1

```

NUMERO DE USUARIOS=          9
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *         *         *         *         *         *
*CPU       *0.3000E-01*0.2326  *0.2915  *0.3759E-01* 7.754  *
*          *         *         *         *         *         *
*DISC  1   *0.1000  *0.3392  *0.4773  *0.1407  * 3.392  *
*          *         *         *         *         *         *
*DISC  2   *0.1000  *0.3392  *0.4773  *0.1407  * 3.392  *
*          *         *         *         *         *         *
*TERMINAL * 8.000   *0.0000E+00* 7.754  * 8.000   *0.9692  *
*          *         *         *         *         *         *
*****
          MEMORY USED:          7578 WORDS OF 4 BYTES
          ( 0.15 % OF TOTAL MEMORY)

R1 = 0.3759E-01
R2 = 0.1407
R3 = 0.1407
X = 0.9692
R = 1.286
N1 = 0.2915
N2 = 0.4773
N3 = 0.4773

```

1

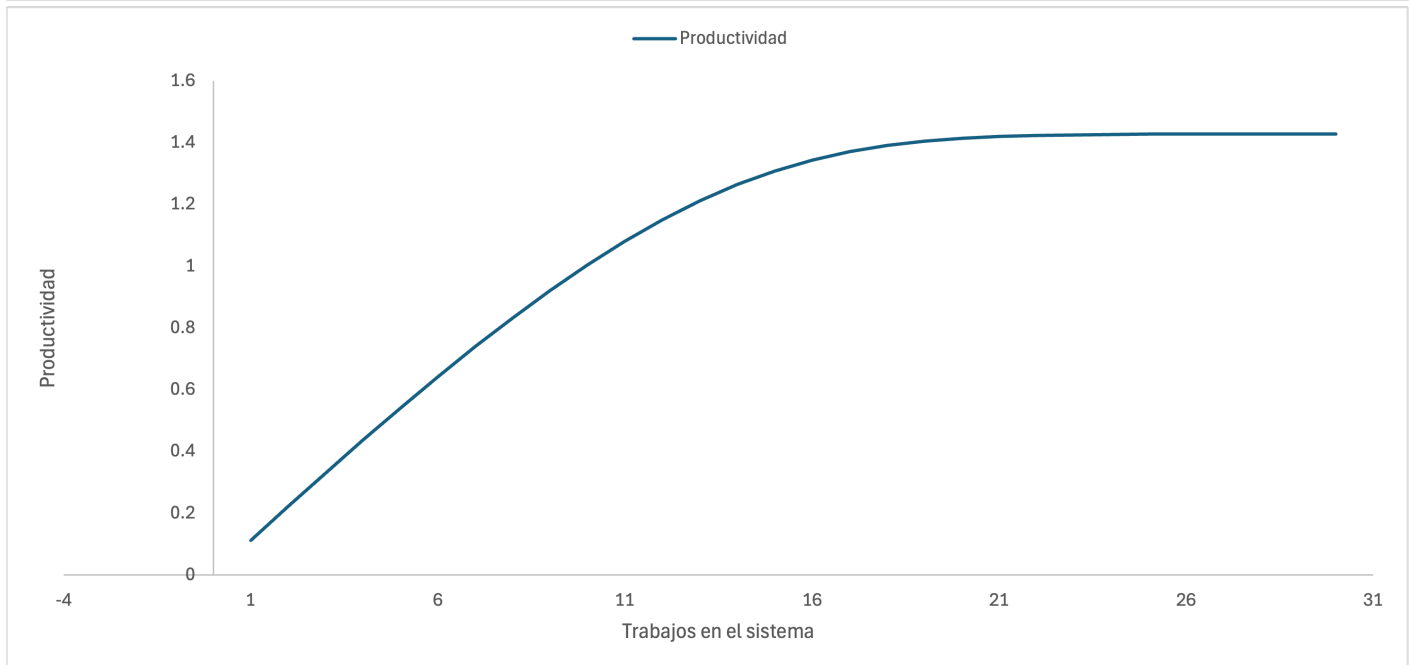
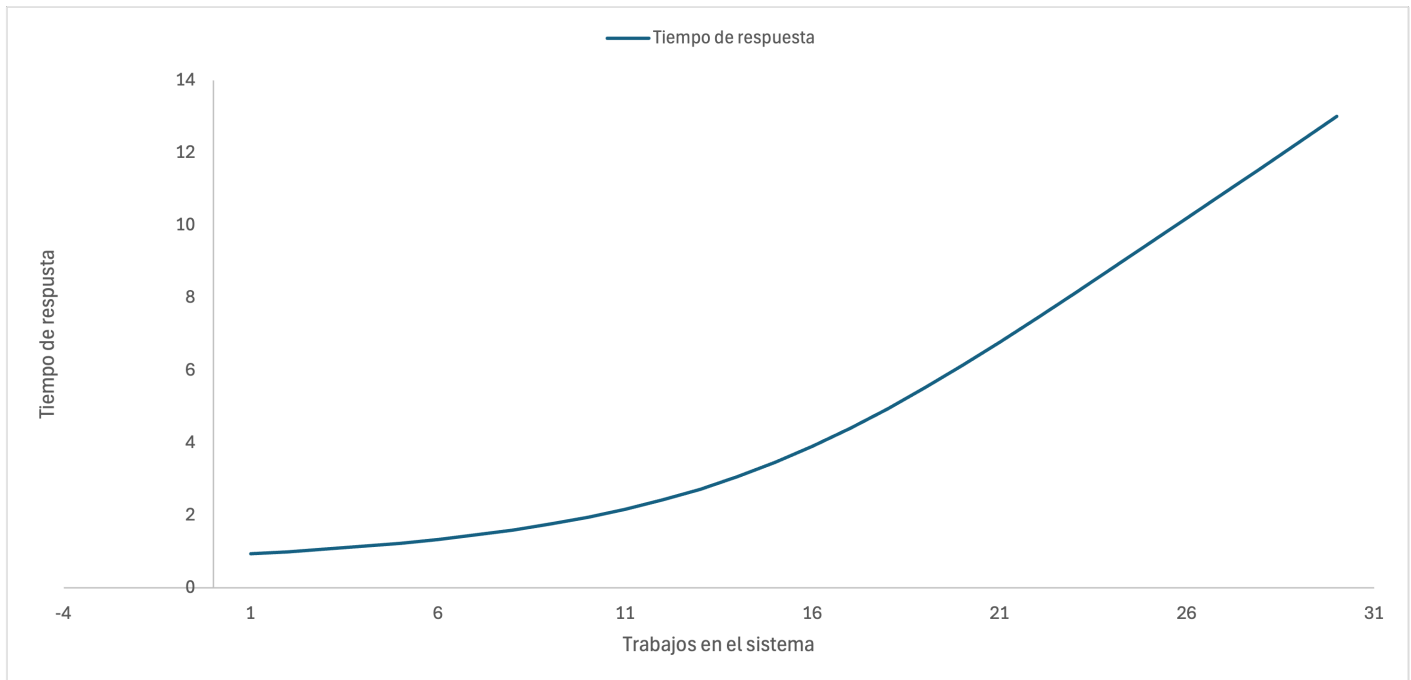
```

NUMERO DE USUARIOS=          10
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *         *         *         *         *         *
*CPU       *0.3000E-01*0.2568  *0.3317  *0.3874E-01* 8.562  *
*          *         *         *         *         *         *
*DISC  1   *0.1000  *0.3746  *0.5534  *0.1477  * 3.746  *
*          *         *         *         *         *         *
*DISC  2   *0.1000  *0.3746  *0.5534  *0.1477  * 3.746  *
*          *         *         *         *         *         *
*TERMINAL * 8.000   *0.0000E+00* 8.562  * 8.000   * 1.070  *
*          *         *         *         *         *         *
*****
          MEMORY USED:          7601 WORDS OF 4 BYTES
          ( 0.15 % OF TOTAL MEMORY)

R1 = 0.3874E-01
R2 = 0.1477
R3 = 0.1477
X = 1.070
R = 1.344
N1 = 0.3317
N2 = 0.5534
N3 = 0.5534
49 /END/

```

5 Apartado d



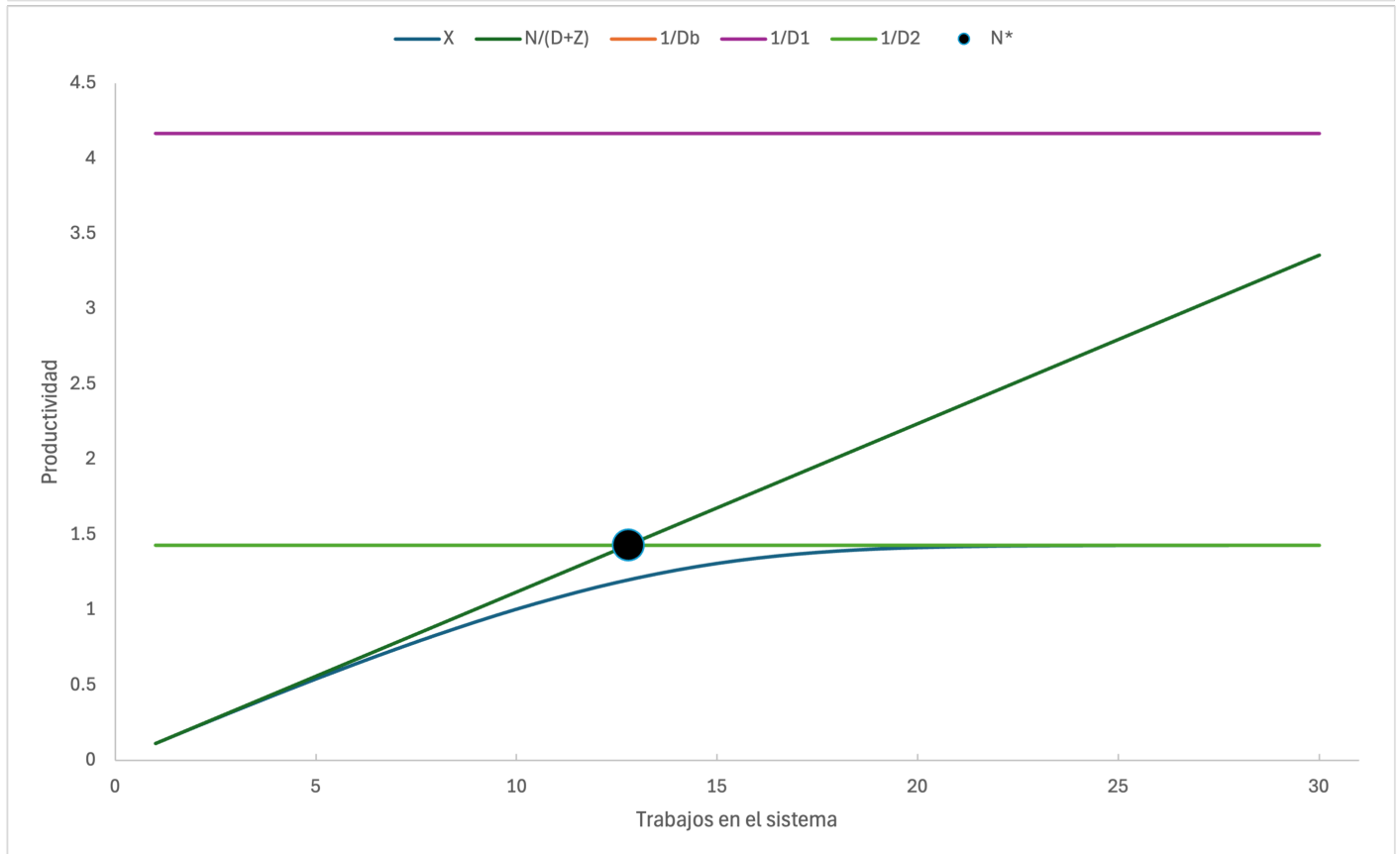
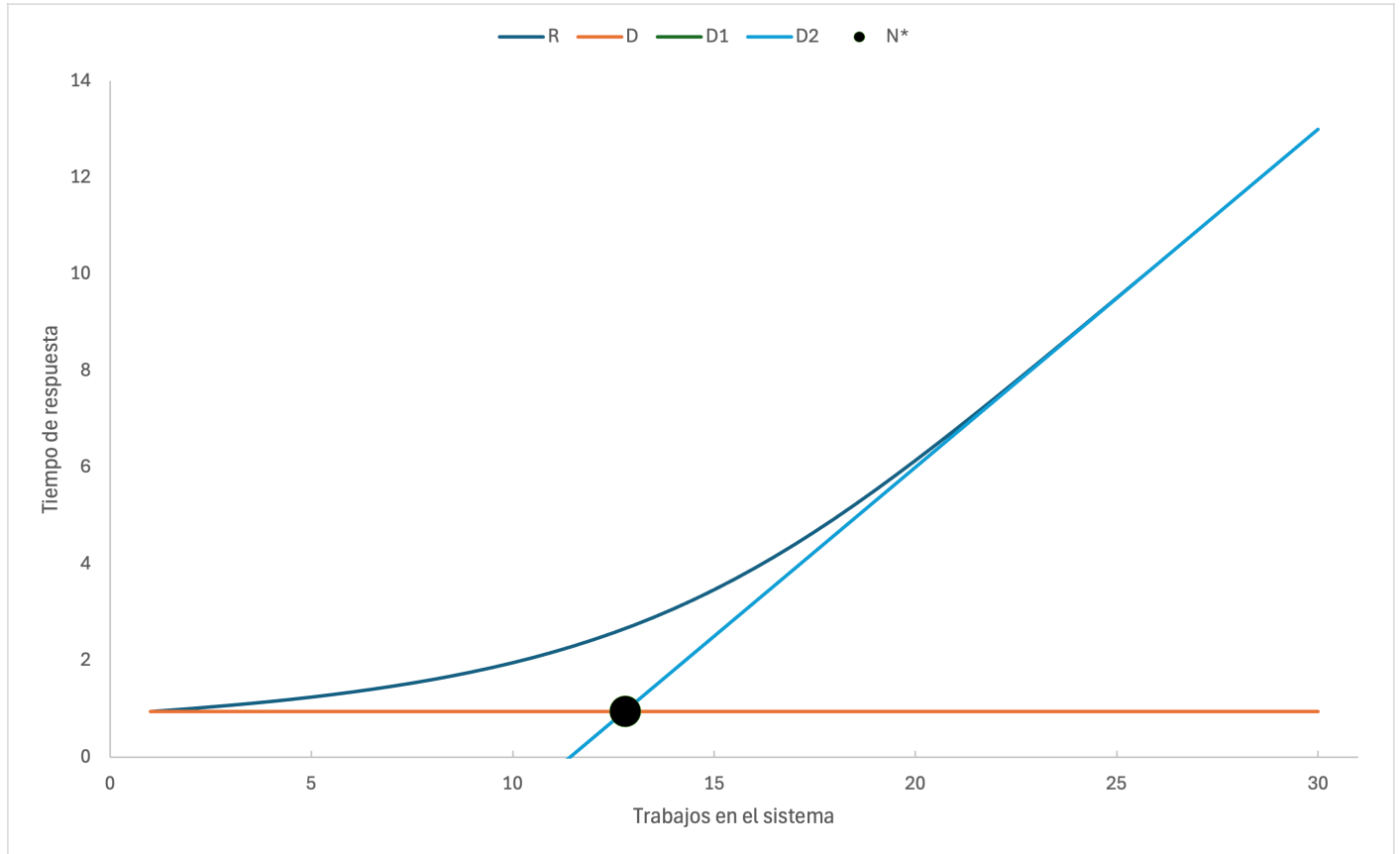
```

1 /DECLARE/ QUEUE CPU,DISC,TERMINAL;
2     REAL Z=8.;
3     REAL V_CPU=8.,V_DISC=7.;
4     REAL X,X1,X2,R,R1,R2,N1,N2;
5     INTEGER N;
6 /STATION/ NAME=CPU;
7 &     SCHED=PS;
8     SERVICE=EXP(0.03);
9     TRANSIT=DISC,V_DISC,TERMINAL,1;
10 /STATION/ NAME=DISC;
11     TRANSIT=CPU;
12     SERVICE=EXP(0.1);
13 /STATION/ NAME=TERMINAL;

```

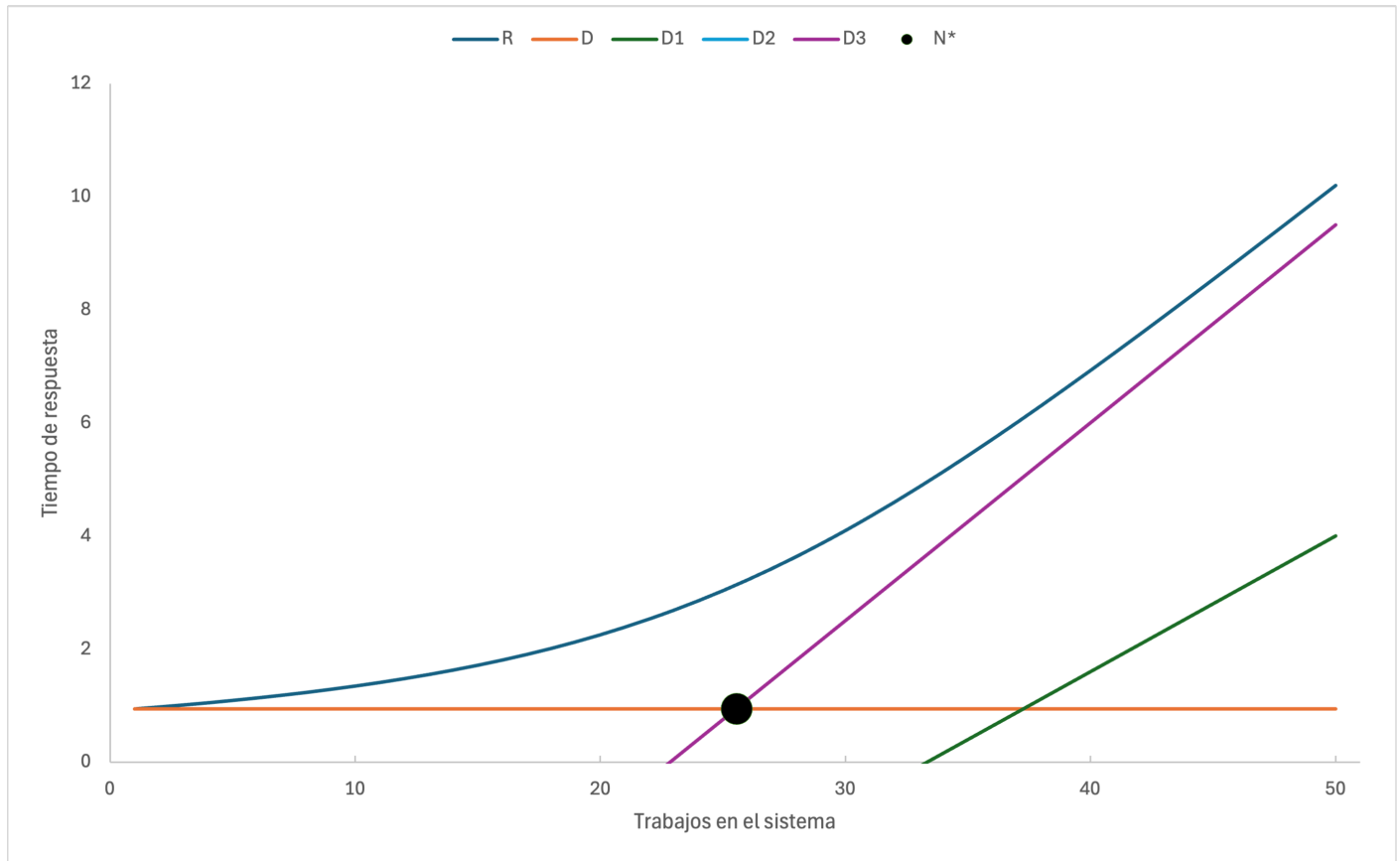
```
14      TYPE=INFINITE;
15      INIT=N;
16      SERVICE=EXP(Z);
17      TRANSIT=CPU;
18 /CONTROL/ CLASS=ALL QUEUE;
19 /EXEC/    FOR N:=1 STEP 1 UNTIL 30 DO
20      BEGIN
21          PRINT;
22          PRINT("NUMERO DE USUARIOS=",N);
23          SOLVE;
24
25          X := MTHRUPUT(TERMINAL);
26          X1 := MTHRUPUT(CPU);
27          X2 := MTHRUPUT(DISC);
28          R := N / X - Z;
29          R1 := MRESPONS(CPU);
30          R2 := MRESPONS(DISC);
31          N1 := X1 * R1;
32          N2 := X2 * R2;
33
34          PRINT("R1 = ", R1);
35          PRINT("R2 = ", R2);
36          PRINT("X = ", X);
37          PRINT("R = ", R);
38          PRINT("N1 = ", N1);
39          PRINT("N2 = ", N2);
40      END;
41 /END/
```

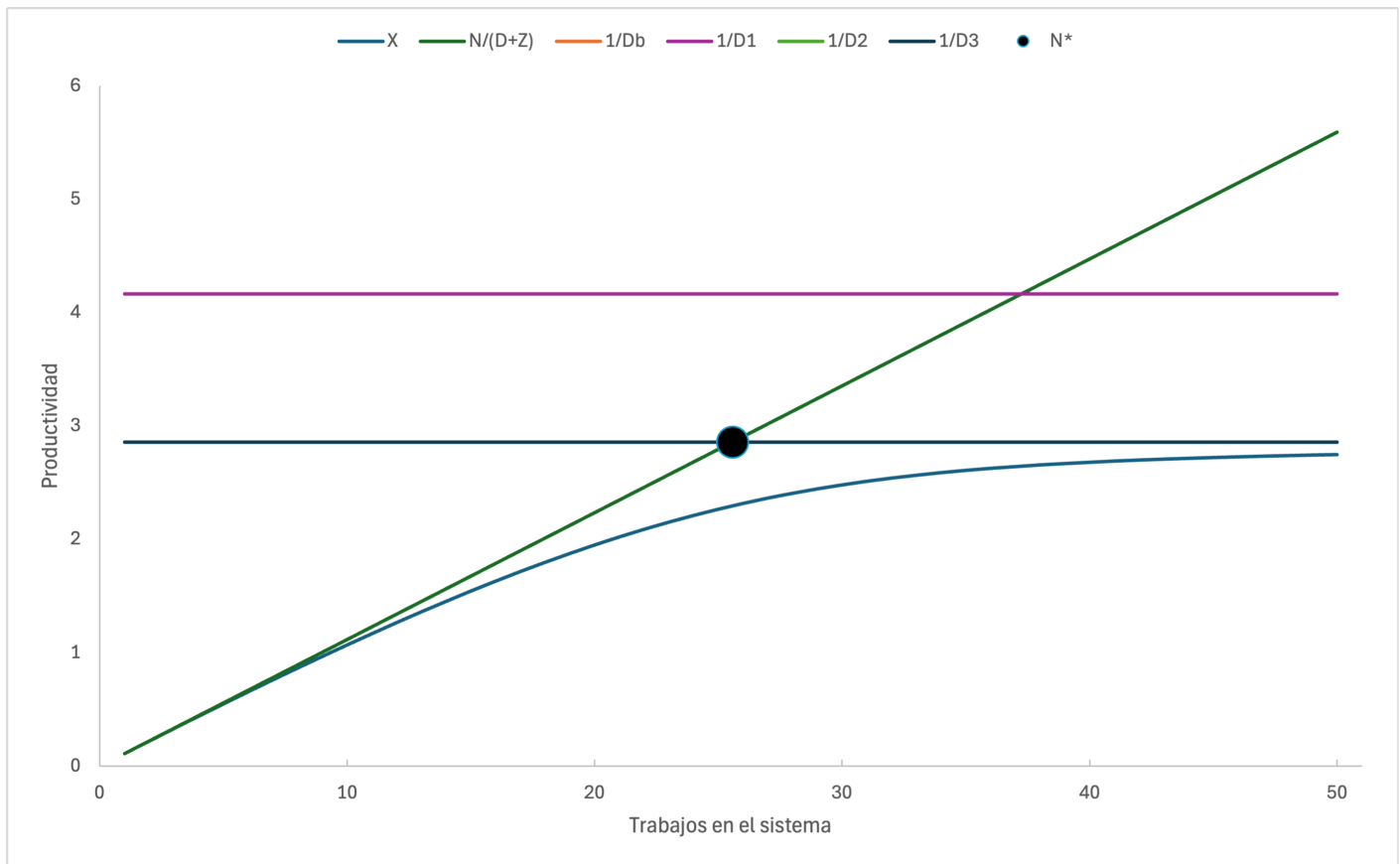
6 Apartado e



7 Apartado f

He incrementado el numero de iteraciones a 50 para poder visualizar mejor la convergencia del tiempo de respuesta con los límites asintóticos.





```

1 /DECLARE/ QUEUE CPU,DISC(2),TERMINAL;
2     REAL Z=8.;
3     REAL V_CPU=8.,V_DISC(2)=(3.5,3.5);
4     REAL X,X1,X2,X3,R,R1,R2,R3,N1,N2,N3;
5     INTEGER N;
6 /STATION/ NAME=CPU;
7 &     SCHED=PS;
8     SERVICE=EXP(0.03);
9     TRANSIT=DISC,V_DISC,TERMINAL,1;
10 /STATION/ NAME=DISC;
11     TRANSIT=CPU;
12 /STATION/ NAME=DISC(1);
13     SERVICE=EXP(0.1);
14 /STATION/ NAME=DISC(2);
15     SERVICE=EXP(0.1);
16 /STATION/ NAME=TERMINAL;
17     TYPE=INFINITE;
18     INIT=N;
19     SERVICE=EXP(Z);
20     TRANSIT=CPU;
21 /CONTROL/ CLASS=ALL QUEUE;
22 /EXEC/   FOR N:=1 STEP 1 UNTIL 50 DO
23     BEGIN
24         PRINT;
25         PRINT("NUMERO DE USUARIOS=",N);
26         SOLVE;
27
28         X := MTHRUPUT(TERMINAL);
29         X1 := MTHRUPUT(CPU);
30         X2 := MTHRUPUT(DISC(1));

```

```
31      X3 := MTHRUPUT(DISC(2));
32      R := N / X - Z;
33      R1 := MRESPONS(CPU);
34      R2 := MRESPONS(DISC(1));
35      R3 := MRESPONS(DISC(2));
36      N1 := X1 * R1;
37      N2 := X2 * R2;
38      N3 := X2 * R3;
39
40      PRINT("R1 = ", R1);
41      PRINT("R2 = ", R2);
42      PRINT("R3 = ", R3);
43      PRINT("X = ", X);
44      PRINT("R = ", R);
45      PRINT("N1 = ", N1);
46      PRINT("N2 = ", N2);
47      PRINT("N3 = ", N3);
48      END;
49 /END/
```